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Predicting diabetes using supervised machine learning algorithms on E-health records

Sulaiman Afolabi [a], Nurudeen Ajadi [b,*], Afeez Jimoh [a], Ibrahim Adenekan [b]

[a] Department of Informatics, University of Louisiana at Lafayette, LA, United States
[b] Department of Mathematics, University of Louisiana at Lafayette, LA, United States

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ABSTRACT

Background: Diabetes mellitus is one of the most significant health challenges currently faced by people especially in the United States of America because of hyperglycemia. Despite recent research on predicting the incidence of the disease, there is still a need for a more efficient and robust approach to accurately predict diabetes, to provide immediate treatment at the early stage.
Methods: This study investigates the early detection and management of diabetes by applying machine learning techniques to electronic health records. The research explores the effectiveness of three supervised machine learning algorithms: logistic regression, Random Forest, and k-nearest neighbors (KNN), in developing predictive models for diabetes. The goal is to identify the most significant features contributing to the disease and to determine which model offers the best performance.
Findings: The KNN model emerged as the top performer among the tested algorithms. It achieved an accuracy of 96.09 %, a sensitivity of 98.54 %, and a specificity of 93.63 %. These results indicate that the KNN model with a mean test error of 0.0391 is the most reliable for predicting diabetes within the studied dataset.
Interpretations: The high sensitivity and specificity suggest that the KNN model is well-suited for distinguishing between diabetic and non-diabetic patients, essential for early diagnosis and effective management of the disease in clinical practice. Despite the dataset's limited demographic scope, the three machine learning algorithms explored, provide useful results.

1. Introduction

Diabetes mellitus, characterized by elevated blood glucose levels (hyperglycemia), is a chronic condition posing significant global health challenges. Projections indicate a concerning trend, with an estimated 642 million individuals worldwide expected to be affected by diabetes by 2040, translating to approximately one in ten adults.[1] The hormone insulin plays a pivotal role in regulating blood glucose levels, and insufficient production or ineffective utilization of insulin are primary factors contributing to diabetes. Left unmanaged, diabetes can lead to a myriad of health complications affecting multiple organs within the body.[2] Alarmingly, the prevalence of diabetes is on the rise, with an estimated 452 million cases identified in 2017 and projected to escalate to 694 million by 2045 (Lawrence e.al., 2021). Moreover, studies suggest that by 2030 and 2045, diabetes prevalence is anticipated to encompass 25 % and 51 % of the population, respectively.[3] The integration of machine learning techniques has significantly influenced various fields, with medicine being a primary beneficiary[4–6]). Leveraging insights derived from extensive data analysis, these technologies facilitate the creation of more precise and personalized approaches to diagnosis and treatment. Consequently, healthcare providers are empowered to make informed decisions promptly, enhancing patient care outcomes.[7] For instance, within oncology, machine learning models can predict treatment responses and customize therapeutic strategies according to each patient's unique characteristics, thereby optimizing treatment efficacy while minimizing adverse effects.

Various algorithms are employed for diabetes prediction, encompassing traditional machine learning methods such as Support Vector Machine (SVM), Decision Tree (DT), and Logistic Regression.[8] Polat & Günes[9] differentiated between diabetes and normal individuals using Principal Component Analysis (PCA) and Neuro-Fuzzy Inference. Recent research on diabetes prediction has seen the widespread adoption of a diverse range of machine learning models, spanning from conventional algorithms like Logistic Regression and k-Nearest Neighbors to

* Corresponding author.
E-mail address: nurudeen.ajadi1@louisiana.edu (N. Ajadi).

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advanced techniques such as Artificial Neural Networks, Random Forests, and Deep Neural Networks. Darolia & Chhillar[10] examined a diabetes dataset utilizing popular algorithms like Artificial Neural Network, Random Forest, and Logistic Regression. Notably, their findings demonstrated that Logistic Regression outperformed other algorithms, underscoring its effectiveness in diabetes prediction. Febrian et al.[11] took a different approach, employing supervised machine learning techniques and comparing two k-Nearest Neighbor algorithms against the Naive Bayes algorithm for diabetes prediction. Interestingly, their study concluded that the Naive Bayes algorithm exhibited superior performance compared to KNN in this context. Mousa et al.[12] conducted a comprehensive comparative study focusing on three widely used models: Long Short-Term Memory, Random Forest, and Convolutional Neural Network for diabetes diagnosis. The article is structured as follows: Section 2 reviews recent literature on the use of algorithms for modeling various healthcare data. Section 3 explains the performance measures and the machine learning algorithms used to model the diabetic data. Section 4 provides an analysis of the compared algorithms. Finally, Section 5 presents the conclusions and recommendations.

2. Literature review

Recent advancement in technology created a world of data that ultimately enables both practical and experimental application of machine learning in health care. Machine learning has been observed to be very useful in diagnosis and prediction of different health challenges whether infectious or non-infectious.[13] Machine learning algorithms make use of AI concepts which are generally classified as supervised, unsupervised and reinforcement learning. The supervised learning approach make use of data to retrieve information from the training set and conceptualize models that can accurately predict the training set outcomes and uses the train model to make predictions of new features in the testing data set.[14] Logistic regression, Decision Trees, Support Vector Machines, Artificial Neural Networks and Random Forest Regression are some of the examples of machine learning algorithms that are based on supervised learning approaches.[15] In an unsupervised learning approach, the algorithms are used to group data into independent clusters which makes it easy for features extraction and classification. In contrast to other forms of approaches in machine learning, unsupervised learning allows for extraction of features by identifying the relationships within the data points and grouping them into clusters based on their similarities.[16] Common examples of unsupervised learning include K-Means, KNN, Deep Belief Network (DBN) and Convolutional Neural Network (CNN).[13] Reinforcement learning techniques are the most efficient form of machine learning algorithms that is closest to human and animal intelligence.[17] This approach works based on self-learning to eliminate error and improve its overall model performance. The most common reinforcement learning method is the Recurrent Neural Network.[13]

Scientists and researchers make use of many machine-learning approaches in diagnosis and prediction of several health challenges to come up with an innovative and better performance solutions compared to the traditional ways of disease diagnosis and treatment in healthcare. They apply the Machine learning algorithms to Electronic Health Records (EHR), Medical Imaging, and genetic engineering to solve different health problems, forecast disease spread and make recommendations about prevention and control mechanisms. El-Bashbishy and El-Bakry[18] proposed a novel technique for early diabetes prediction with high accuracy. They optimized data preprocessing, prediction, and classification using a novel dataset of Mansoura University Children's Hospital Diabetes (MUCHD), which allowed for a comprehensive evaluation of the system's performance. Various validation metrics were employed to ensure the reliability of the results using cross-validation approaches with various statistical measures of accuracy, F-score, precision, sensitivity, specificity, and Dice similarity coefficient.[19] introduced the first-ever self-explanatory interface for diagnosing diabetes patients using machine learning. They proposed four classification models (Decision Tree (DT), K-nearest Neighbor (KNN), Support Vector Classification (SVC), and Extreme Gradient Boosting (XGB)) based on the publicly available diabetes dataset. All the models exhibited commendable accuracy in diagnosing patients with diabetes, with the XGB model showing a slight edge over the others.

Liu et al.[20] developed a deep learning algorithm model using a combination of reinforcement and supervised learning approaches to accurately predict the beginning of various common diseases such as stroke, kidney failure, and heart failure. The prediction model makes use of both structured and unstructured data from EHR and diagnosis notes which resulted in the model high performance and versatility. In another research using EHR data, Ahmad and Ali[21] predict mortality in paralytic ileus (PI) - a medical condition characterized by incomplete blockage of the intestine that prevent direct passage of food substance and which may later lead to total blockage of the intestine. In this research, they come up with an algorithm that predict the mortality in PI patients with an accuracy of 81.3 %. This development leads to improved awareness as well as robust clinical treatment for the ailment.

Medical imaging is another important health field that has and continues to witness the transformative application of machine learning applications in healthcare. Machine learning applications and techniques have been employed to improve imaging modalities in areas like Computed Tomography (CT), Ultrasound, X-Ray, Magnetic Resonance Imaging (MRI) to mention a few.[3] With Medical imaging, McKinney et al.[22] recently developed a deep learning algorithm to detect tumors in Mammogram images in the early stage of cancer. A careful comparison to assess the performance of this algorithm shows that it outperforms an experienced radiologist who use the traditional method of tumors detection by 11.5 %. In the same vein, Esteva et al.[23] make use of a convolutional neural network for a classification of 2032 different skin diseases with dermoscopic images. The performance of the Classification algorithms compares with practical method by 21-board member of registered dermatologist give similar results. Diabetic Retinography is a common eye condition that usually affects up to 60 % of type 1 diabetic patient, which is difficult to detect at the early stage.[24] Alqudah[25] applied a deep learning CNN to detect the aneurysms that cause impaired eye vision in diabetic retinography progression. The early detection of this abnormality through the CNN approach makes it possible to prevent irreversible damage to the patient's vision.

In genetic engineering, machine learning applications, especially the deep learning techniques have tremendously been applied in solving most of the health challenges that seem not treatable. For example, in pharmacogenomics, machine learning procedures are instrumental in the determination of stable doses of medication for patients to avoid unwanted reactions after treatment.[26] Tang et al.[27] applied a machine learning based algorithm to determine a stable dose of an immunosuppressive medication for patients who underwent procedures for a renal transplant to reduce the risk of rejection. Further application of machine learning in this field has been witnessed in psychiatry, oncology, and neurology.[28]

Finally, the importance of machine learning and its application in genetic engineering also extends to the development of vaccines to fight against the COVID 19 pandemic. Malone et al.[29] make use of machine learning based software to predict which antigen contain the necessary features of HLA binding, processing, and the potential to be recognized by T-cells to be a useful target for immunotherapy. The application of immunogenicity from the software-based algorithm with other medical considerations make it possible to formulate a universal vaccine that can be used to combat the covid19 pandemic.

This research illuminates the diverse strengths of these models and offers valuable insights into their suitability for diabetes prediction.[26] Despite the existing literature on diabetes prediction, this study aims to contribute to the field by predicting the likelihood of developing diabetes using machine learning techniques. For more articles on the application of machine learning to diabetes, read,[37–39] Yang et al.[40] and Salman Shukur et al.,.[41]

3. Research methodology

In this section, the supervised machine learning algorithms used in diabetes prediction were explained. The algorithms include K – Nearest Neighbors (KNN) Algorithm, Random Forest (RF) Algorithm and Logistic Regression. In this article, we explored some of the properties of these algorithms.

3.1. K – nearest neighbors (KNN) algorithm

The K – Nearest Neighbors (KNN) algorithm is a supervised machine learning method employed to tackle classification and regression problems. Fix & Hodges[30] developed this algorithm in 1951, which was subsequently expanded by Thomas Cover.[31]

The K – Nearest Neighbors (KNN) algorithm is a widely utilized and straightforward machine learning technique, prized for its simplicity and ease of implementation. Notably, it operates without requiring any assumptions regarding the underlying data distribution, thereby accommodating both numerical and categorical data. This inherent flexibility renders KNN a suitable choice for various classification and regression tasks. As a non-parametric method, KNN bases its predictions on the similarity between data points, exhibiting reduced sensitivity to outliers compared to other algorithms.[32]

The KNN algorithm functions by identifying the K nearest neighbors to a given data point based on a chosen distance metric, such as Euclidean distance. The class or value of the target data point is then determined by either the majority vote or the average of these K neighbors. This methodology allows the algorithm to adapt to varying data patterns and make predictions grounded in the local structure of the dataset.[33]

A mathematical representation of the k-Nearest Neighbors (KNN) algorithm:

The most used distance metric in KNN is the Euclidean distance, which is defined as:

$$d(x,y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

where x and y are two data points in an n-dimensional space, and x_i and y_i are the ith features of x and y, respectively.

Prediction for Classification: The predicted class $\hat{y}$ for a data point x is determined by the majority vote among its k-nearest neighbors. Mathematically, this can be represented as:

$$\hat{y} = \underset{c \in C}{argmax} \sum_{i=1}^{k} \mathbb{I}(\quad y_i = c)$$

where C is the set of possible classes, y_i is the class label of the ith nearest neighbor, and $\mathbb{I}$ is the indicator function that returns 1 if the argument is true and 0 otherwise.

Prediction for Regression: The predicted value $\hat{y}$ for a data point x is the average of the values of its k-nearest neighbors. Mathematically, this is represented as:

$$\hat{y} = \frac{1}{k} \sum_{i=1}^{k} y_i$$

where y_i is the value of the ith nearest neighbor.

3.2. Random forest (RF) algorithm

A Random Forest algorithm is a supervised machine learning algorithm that is extremely popular and is used for classification and regression problems. It is based on the concept of ensemble learning, which is a process of combining multiple classifiers to solve a complex problem and to improve the performance of the model. Heath et al.,[34] first proposed the method of random decision forests which was further developed by Ho.[35]

The Random Forest algorithm is a highly effective machine learning technique that constructs multiple decision trees during the training phase. Each tree is built using a random subset of the dataset and considers a random subset of features at each split. This randomness introduces variability among the individual trees, reducing the risk of overfitting and enhancing overall prediction performance. For predictions, the algorithm aggregates the outputs of all the trees, either through majority voting (for classification tasks) or averaging (for regression tasks). This collective decision-making process, supported by numerous trees, yields stable and accurate results. Random forests are widely used for both classification and regression tasks, known for their ability to handle complex data, minimize overfitting, and provide reliable predictions across various applications.[36]

The Random Forest consist of an ensemble of B decision trees. For a given input x, the b-th tree in the forest outputs a prediction $h_b(x)$.

For regression tasks, the final prediction $\hat{y}$ is the average of predictions from all trees:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} h_b(x)$$

For classification tasks, the final prediction $\hat{y}$ is based on majority voting:

$$\hat{y} = mode\{h_1(x), h_2(x), \ldots, h_B(x)\}$$

3.3. Logistic regression in machine learning

Logistic Regression is used in predicting the probability of classifying an item. It is an example of a supervised learning algorithm. Binary Logistic Regression classified an item into two(binary) using sigmoid function.

A Logistic Regression is of the form:

$$\log\left(\frac{Y}{1-Y}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p$$

By taking the exponent, the Logistic Regression is written as:

$$P(Y/X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p)}}$$

Where X is the independent variable and Y is dependent variable (binary)

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1m} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nm} \end{bmatrix}$$

$$Y = \begin{cases} 0 if\ item1 \\ 1 if\ item2 \end{cases}$$

The Logistic Regression is S-shaped and fitting the S-shaped function helps in predicting the outcome of two values.

3.4. Evaluation metrics

When developing and evaluating classification models, it is essential to consider multiple metrics that reflect different dimensions of the model's performance. These metrics collectively offer a thorough evaluation of how effectively the model is working for a given task. The following criteria are key to understanding a classification model's overall performance:

3.4.1. Accuracy

Accuracy is a fundamental metric used to evaluate the overall correctness of a model. It is the ratio of correctly predicted instances to the

total number of instances in the dataset. It refers to the percentage of all instances that have been predicted correctly i.e it indicates how often the model makes correct predictions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

3.4.2. Sensitivity

Sensitivity is a key metric for evaluating a model's ability to correctly identify positive instances. It is the proportion of true positives (TP) out of all actual positive instances in the dataset.

$$Sensitivity = \frac{TP}{TP + FN}$$

3.4.3. Specificity

Specificity is a critical metric used to evaluate a model's ability to correctly identify negative instances within a dataset. It is very important in applications where the cost of false positives is substantial, such as in medical diagnostics, fraud detection, or security systems to avoid false alarms. It is defined as the proportion of true negatives (TN) out of all actual negative instances.

$$Specificity = \frac{TN}{TN + FP}$$

Where:

TP (True Positives): The number of instances where the model correctly predicts the positive class.

TN (True Negatives): The number of instances where the model correctly predicts the negative class.

FP (False Positives): The number of instances where the model incorrectly predicts the positive class.

FN (False Negatives): The number of instances where the model incorrectly predicts the negative class.

4. Data analysis

In this study, we analyzed the diabetic data by comparing efficiency of the three supervised machine learning algorithms. The evaluation metrics used include Accuracy, Specificity and Sensitivity.

4.1. The dataset description

The Diabetes dataset is gotten from the United States Center for Disease and Control database to show the categories of people that are either positive or negative to diabetes test. The dataset contains different variables that directly or indirectly lead to diabetes and some symptoms variable that happen when someone is positive to the diabetes test. The dataset contains 100000 records with 9 variables. The dataset can be access via this link Diabetes dataset. The participants in this study range in age from as young as a one-month-old baby to 80 years old. Likewise, the body mass index (BMI) of participants spans from a minimum of 10.01 to a maximum of 95.69. This demonstrates the diverse and extensive range of the data used for the analysis. The dataset contains three data types which are the character (chr), numeric (num) and integer (int) data. This is crucial for statistical analysis because it allows for appropriate data handling, guides model selection, and improves performance and accuracy of the model. The variable description is presented in Table 4.1.

4.1.1. Data cleaning

Data cleaning involved checking for duplicate entries within the dataset, revealing that over 2500 observations were duplicated. The subset and negation functions were used to remove these duplicates and verify the cleaned dataset. An essential aspect of data cleaning is checking for missing or null values, as unaddressed missing data can lead to improper model development and often cause overfitting, which is detrimental to creating a realistic and high-performance machine learning model.

Table 4.1
Variable Description.

Variable (column) name	Mode	Description
Gender	character	This pertains to an individual's biological sex.
Age	number	Age is a significant factor since research has shown that older individuals are generally more susceptible to diabetes.
Hypertension	number	This is a medical condition in which blood pressure consistently remains above normal levels.
Heart_disease	number	This is another medical condition that is associated with an increase of developing diabetes.
Smoking_history	character	This is a risk factor that is associated with diabetes and can also worsen the complication with the disease.
Body Mass Index (Bmi)	number	This is a measure of body fat based on weight and height. Usually, higher bmi values correspond to higher risk of diabetes
Hemoglobin (HbA1c_level)	number	This is a measure of a person's average sugar level over the past 2–3 months. In most cases, more than 6.5 % HbA1c_level indicates a positive test to diabetes.
Blood_glucose_level	integer	This refers to the amount of glucose in the bloodstream at a given time.
diabetes	integer	This is the main target variable being predicted which is a medical condition that either shows your body system doesn't have enough insulin or cannot convert the insulin for food break down

4.2. Data exploration and analysis

The data exploration was used to gain insights into the distribution of age-group of people who either test negative or positive to diabetes. The plot in Fig. 4.1 shows that older participants are more diagnosed with diabetes, this may be due to the low immune system. Also, in Fig. 4.2, the female population are higher than that of male, but each sex has the almost equal number of people who are affected by the chronic ailment (diabetes). What can be inferred from this plot is that diabetes is not gender biased as it does not give priority to the sex of individual.

The descriptive statistics of the train and test datasets are presented in Tables 4.2 and 4.3. The datasets were classified as group1 for non-diabetic and group2 as diabetic. It is observed that there is older age in diabetic group. The diabetic group has a much higher mean age (82.98 years) compared to the non-diabetic group (61.48 years). This suggests that diabetes is more prevalent or detected at older ages in this dataset. The standard deviation is smaller for the diabetic group (14.54 vs. 22.83), indicating that the ages of diabetic individuals are more concentrated around the mean. This could suggest that older individuals in this dataset have a higher likelihood of being diagnosed with diabetes, leading to a tighter age range.

4.2.1. Correlation plot of numerical variables

The correlation of all variables was examined to understand their relationships within the dataset. Fig. 4.3 indicates that while the numeric variables are not strongly correlated, they all exhibit positive correlation coefficients with each other. The strongest correlation is between age and body mass index (BMI), with a correlation coefficient of 0.3447791. This suggests that older individuals tend to have higher BMI values compared to younger ones.

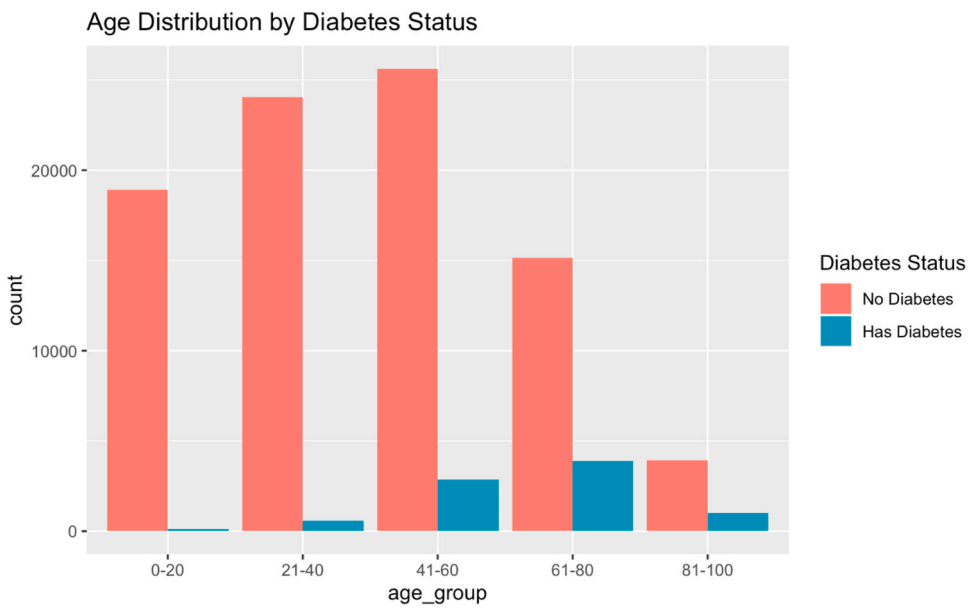

Fig. 4.1. Age Distribution by diabetes status.

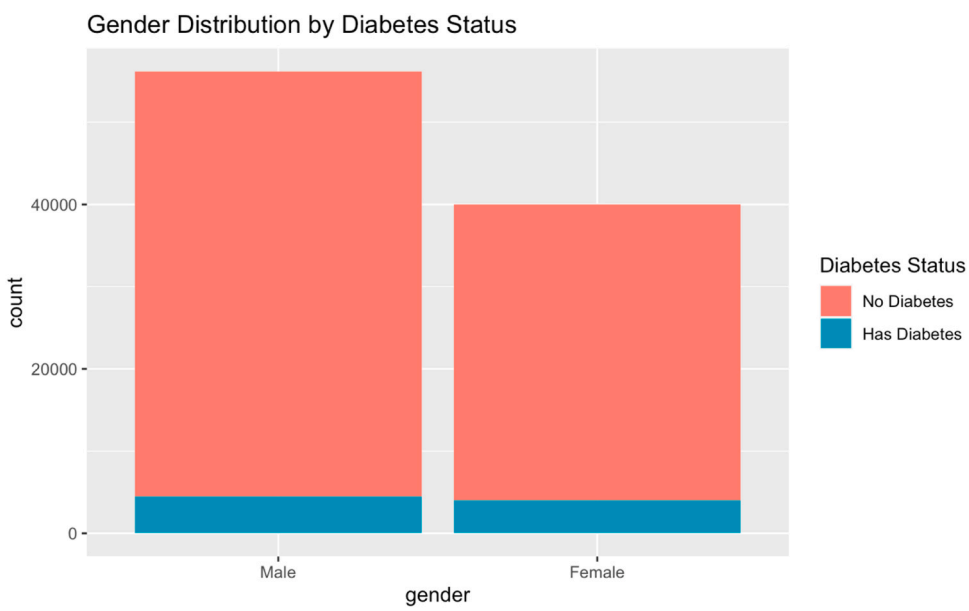

Fig. 4.2. Gender Distribution by diabetes status.

Table 4.2
Descriptive statistics of the Train datasets.

item	n	Mean	SD	Min	Max	Skew	Kurt	S.E
age1	34004	39.60787	22.28	0.08	80	0.0441	−0.9773	0.12084
age2	33285	61.04248	14.51	3	80	−0.764	0.44881	0.07951
bmi1	34004	26.82111	6.481	10.01	95.22	0.8714	2.65611	0.03515
bmi2	33285	32.02948	7.645	10.98	88.72	1.2555	2.69043	0.0419
HbA1c_level1	34004	5.392145	0.972	3.5	6.6	−0.549	−0.954	0.00527
HbA1c_level2	33285	6.945068	1.081	5.7	9	0.7332	−0.8099	0.00592
blood_glucose_level1	34004	132.7565	34.3	80	200	−0.005	−0.8035	0.18602
blood_glucose_level2	33285	194.1429	58.52	126	300	0.502	−1.2305	0.32075
hypertension1	34004	0.06161	0.24	0	1	3.6463	11.2958	0.0013
hypertension2	33285	0.24858	0.432	0	1	1.1634	−0.6465	0.00237
heart_disease1	34004	0.028644	0.167	0	1	5.6514	29.9393	0.0009
heart_disease2	33285	0.148686	0.356	0	1	1.9748	1.8999	0.002

Table 4.3
Descriptive statistics of the Test datasets.

item	n	Mean	SD	Min	Max	Skew	Kurt	S.E
age1	14355	40.05874	22.33	0.08	80	0.0401	−0.9821	0.1864
age2	14484	60.84334	14.62	3	80	−0.78	0.50443	0.12151
bmi1	14355	26.89559	6.555	10.21	95.22	1.0461	3.70059	0.05471
bmi2	14484	32.13336	7.749	11.95	88.72	1.3144	2.89444	0.06439
HbA1c_level1	14355	5.400613	0.968	3.5	6.6	−0.567	−0.9252	0.00808
HbA1c_level2	14484	6.920202	1.075	5.7	9	0.7762	−0.7337	0.00893
blood_glucose_level1	14355	133.2325	34.27	80	200	−0.013	−0.7867	0.28607
blood_glucose_level2	14484	194.3828	58.41	126	300	0.4976	−1.2222	0.48535
hypertension1	14355	0.06193	0.241	0	1	3.6346	11.2114	0.00201
hypertension2	14484	0.244822	0.43	0	1	1.1868	−0.5915	0.00357
heart_disease1	14355	0.032323	0.177	0	1	5.2882	25.9669	0.00148
heart_disease2	14484	0.149544	0.357	0	1	1.9652	1.86215	0.00296

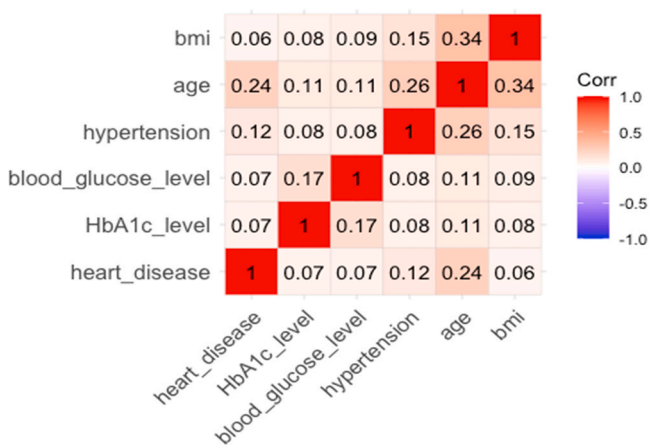

Fig. 4.3. Correlation plot of variables.

4.3. *Model evaluation and performance evaluation*

In this article, three machine learning algorithms were utilized for performance evaluation: logistic regression (LR), k-nearest neighbors (KNN), and random forest (RF). The logistic regression model uses the diabetes variable as the predictor and other variables as explanatory variables. Various subset selections were performed to validate and finalize the model's performance using a validation approach.

As shown in Table 4.4, most variables in the dataset are statistically significant concerning diabetes, although some are not. Variables such as age, gender, BMI, hypertension, heart disease, HbA1c level, and blood glucose have very low p-values below 0.05, indicating strong relationships. Further analysis involved creating a predictive model and validating it using error rate to assess performance and accuracy.

The model's accuracy, sensitivity, and specificity are presented in Table 4.6. The optimal tuning parameters in Table 4.5 for KNN are K= 3 and Mtry= 4, which aid in parameter optimization. In training the machine learning algorithm, 70 % of the data was used for training, and the remaining 30 % was used for testing.

For the logistic regression (LR) model, the prediction results showed an error rate of 11.64 %, with an accuracy of 88.36 %, sensitivity of 87.88 %, and specificity of 88.83 %. Using the grid search method for varying values of k (3, 4, 5, 6), the tuning parameter k = 3 was selected to evaluate the KNN model's performance. The results indicated an accuracy of 88.36 %, sensitivity of 87.88 %, and specificity of 88.83 %. A similar approach was applied to the hyperparameters of the RF classifier, with Mtry= 4 as the tuning parameter, resulting in an accuracy of 88.36 %, sensitivity of 87.88 %, and specificity of 88.83 %.

These results demonstrate that the KNN model is the best predictive model for the diabetes dataset used in this study due to its high performance.

Based on the SHAP analysis depicted in the figure, it is evident that the age variable exerts the greatest positive significance on the model's prediction for diabetes, compared to the other variables. This suggests

Table 4.5
Chosen parameters optimized for each classifier.

Models	Optimal tuning parameter	Mean test error
Logistics Regression	None	0.1164
KNN	K= 3	0.0391
Random Forest	Mtry= 4	0.0536

Table 4.6
Evaluation metrics for the dataset.

Models	Accuracy	Sensitivity	Specificity
Logistic Regression	88.36 %	87.88 %	88.83 %
KNN	96.09 %	98.54 %	93.63 %
Random Forest	94.64 %	96.01 %	93.25 %

Table 4.4
Logistic Regression Model.

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	−2.393e+ 01	2.254e−01	−106.169	< 2e−16 * **
genderMale	3.263e−01	2.853e−02	11.437	< 2e−16 * **
age	5.126e−02	8.605e−04	59.570	< 2e−16 * **
hypertension	7.919e−01	4.055e−02	19.529	< 2e−16 * **
heart_disease	7.269e−01	5.480e−02	13.263	< 2e−16 * **
smoking_historyever	1.352e−01	7.327e−02	1.846	0.0649.
smoking_historyformer	−2.249e−02	5.558e−02	−0.405	0.6857
smoking_historynever	−4.287e−02	4.683e−02	−0.915	0.3600
smoking_historyNo Info	−5.403e−01	5.074e−02–10.649	< 2e−16 * **	
smoking_historynot current −7.996e−02	6.605e−02	−1.211	0.2260	
bmi	9.536e−02	2.157e−03	44.208	< 2e−16 * **
HbA1c_level	2.164e+ 00	2.690e−02	80.427	< 2e−16 * **
blood_glucose_level	3.143e−02	4.207e−04	74.710	< 2e−16 * **

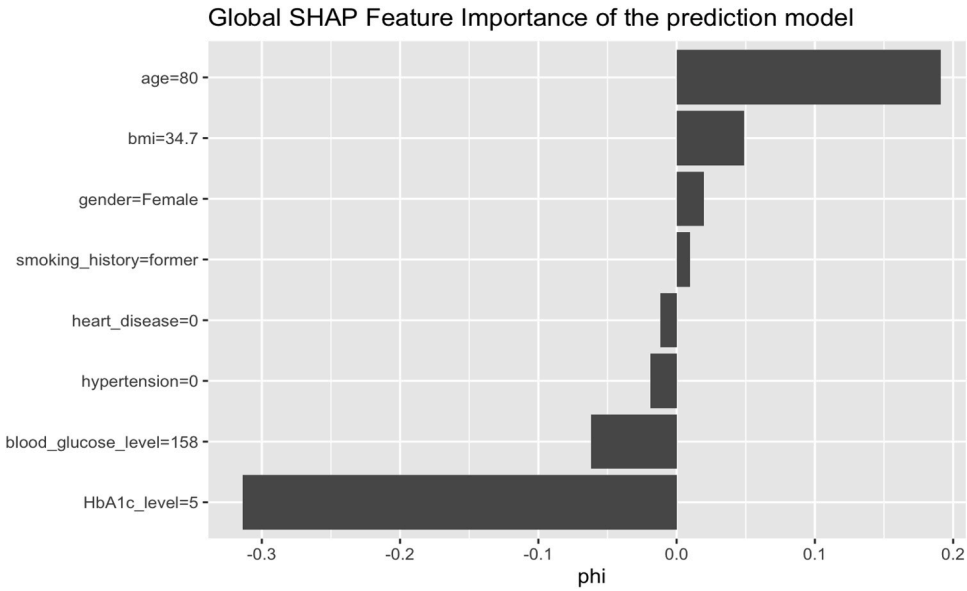

Fig. 4.4. Global SHAP feature for the prediction model.

that being 80 years old significantly increases the likelihood of the model predicting diabetes for the patients. Meanwhile, the HbA1c level demonstrates the most substantial negative influence on the model's prediction. As indicated in the plot, a HbA1c level of 5 % contributes to a lower probability of predicting diabetes, reflecting the importance of controlled blood glucose levels in reducing diabetes risk.

5. Discussion

This study utilized the diabetes dataset from the United States Center for Disease Control database to analyze various factors associated with diabetes focusing on exploratory data and building baseline machine learning models. Till today, an effective cure (medication) or vaccine for the ailment in the health industry remains a mystery. As a result, it is imperative to devise different ways or methods of managing patients diagnosed with diabetes to reduce the fatality rate as well as promote preventive measures of contracting the illness.

Our research focuses on predicting diabetes and understanding major features that contribute to the ailment through the application of three (3) machine learning algorithms (logistic regression, Random Forest, and KNN) which are used to develop models and make predictions. Several model evaluations and parameters were used for the model assessment to come up with the best-performing models and models that offer the highest reliability in terms of accuracy, minimal test error, specificity, and sensitivity. We used a dataset with 100,000 records and nine variables related to diabetes, such as age, gender, BMI, hypertension, heart disease, smoking history, HbA1c levels, blood glucose levels, and whether the person had diabetes.

From the analysis of the result, Older age groups exhibited a higher prevalence of diabetes, potentially due to weakened immune systems. Also, Gender did not significantly influence diabetes risk, suggesting it's not gender-biased. While individual numeric variables lacked strong correlations, they generally showed a positive correlation with each other. Notably, age had the strongest correlation with body mass index (BMI), indicating a tendency for older individuals to have higher BMIs.

Furthermore, the different machine learning approaches on the data set were evaluated, it is evident that KNN has the best performance model with the highest accuracy and performance among the three-machine learning adopted for this research. So, we recommend that for best disease prediction in machine learning, the KNN algorithm offers a more robust result that can be relied upon when tackling diabetes with this dataset.

5.1. Conclusion

To conclude, the main purpose of this study is to contribute to the continuous research of finding a lasting solution or probably a cure to diabetes which remains adamant today. Scientists and researchers from government and non-governmental organizations continue to explore different technologies and methods to combat the disease which has resulted in the development of various medications and vaccines to reduce the effect on patients. On our end, we believe that the application of machine learning algorithms on electronic health data records will go a long way in the early discovery of the disease as well as implementing a lasting solution to it.

5.2. Recommendation

It is noteworthy to mention that real-life monitoring of patient data in clinical research will be a valuable step towards reimagining diabetes studies as it will provide more accurate data that machine learning engineers and scientists can depend on for successful and impactful analysis, modeling, and predictions.

The study, while insightful, has some limitations that should be considered. The dataset was obtained from the United States Centre for Disease Control and may not fully represent the global population. This dataset's limited demographic scope could affect the generalization of the results. Additionally, only three machine learning algorithms were explored, providing useful results. However, more advanced techniques like deep learning can be considered to capture complex patterns more effectively.

For future study, this research could be enhanced using more advanced machine learning techniques and extension of machine learning algorithms beyond the three adopted for this research. Advanced and deep learning algorithms can also be used to investigate the trends in disease as well as the development of models and predictions that are more robust and reliable for the success of future research.

CRediT authorship contribution statement

Afeez Jimoh: Writing – review & editing, Writing – original draft, Investigation. **Ibrahim Adenekan:** Writing – review & editing, Visualization, Methodology. **Sulaiman Afolabi:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation,

Conceptualization. **Nurudeen Ajadi:** Writing – review & editing, Supervision, Methodology, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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